

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

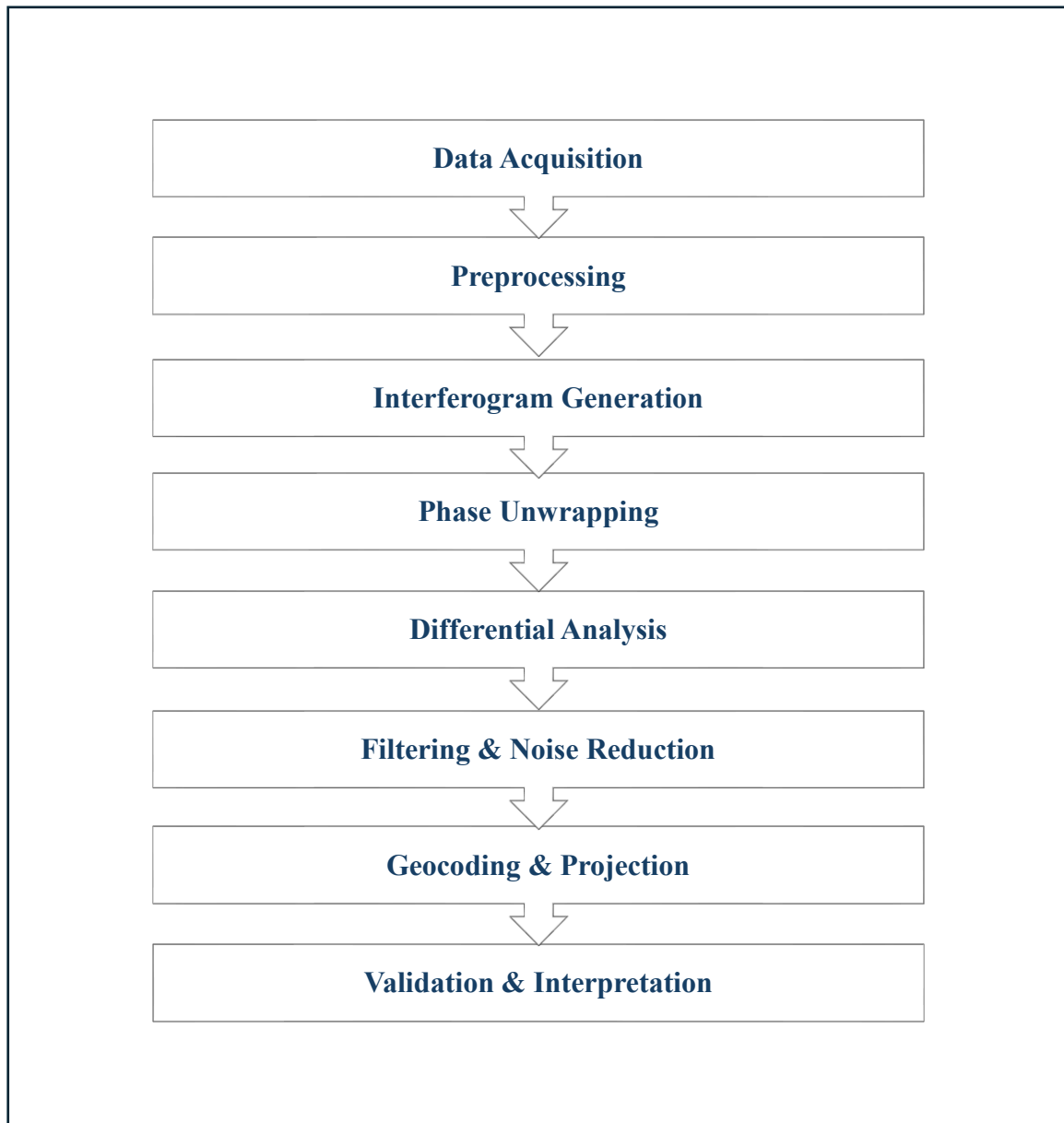
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 946+350

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

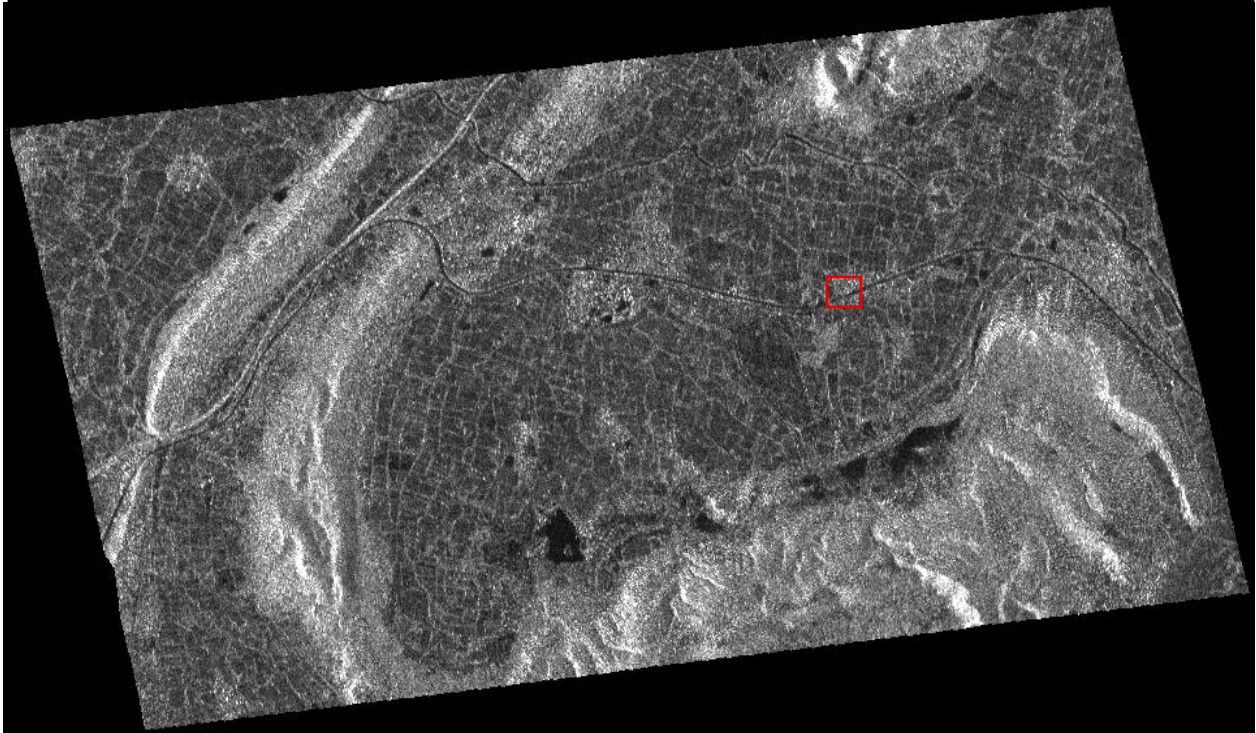
Flight Direction - DESCENDING

Polarization – VV+VH

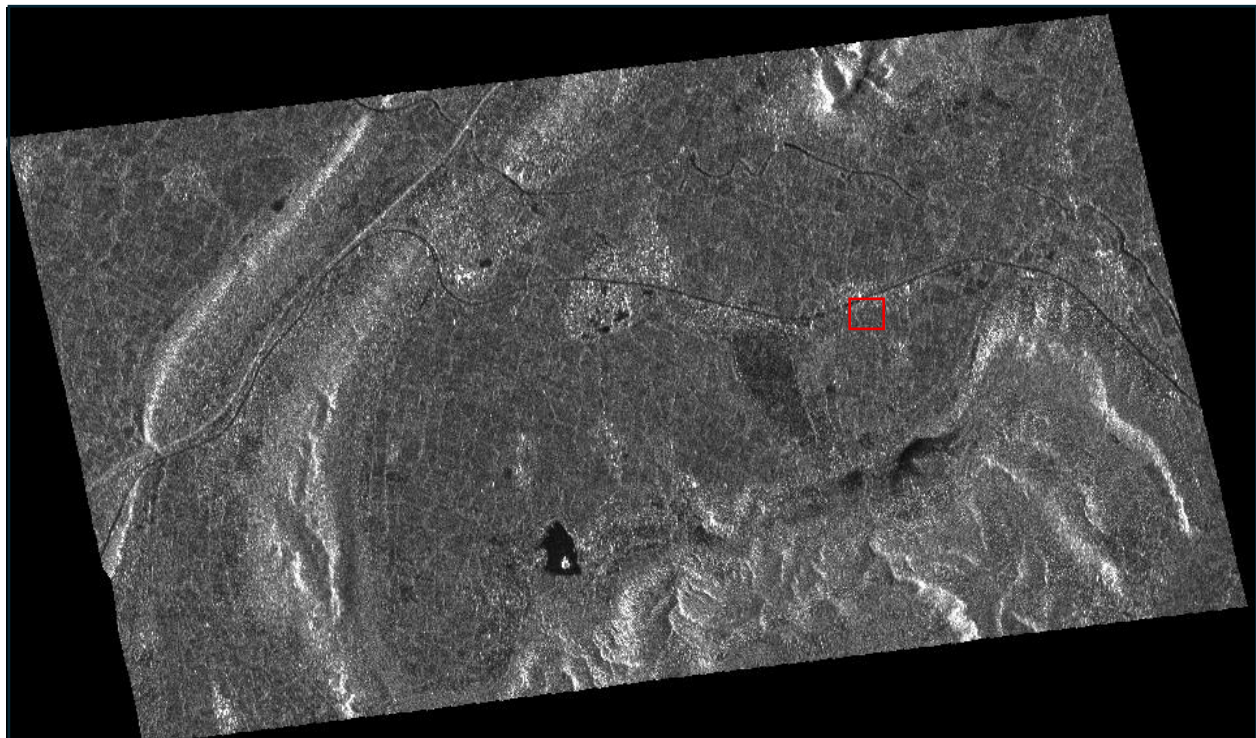
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orth rectification of SAR image VH Polarization



- Orth rectification of SAR image VV Polarization

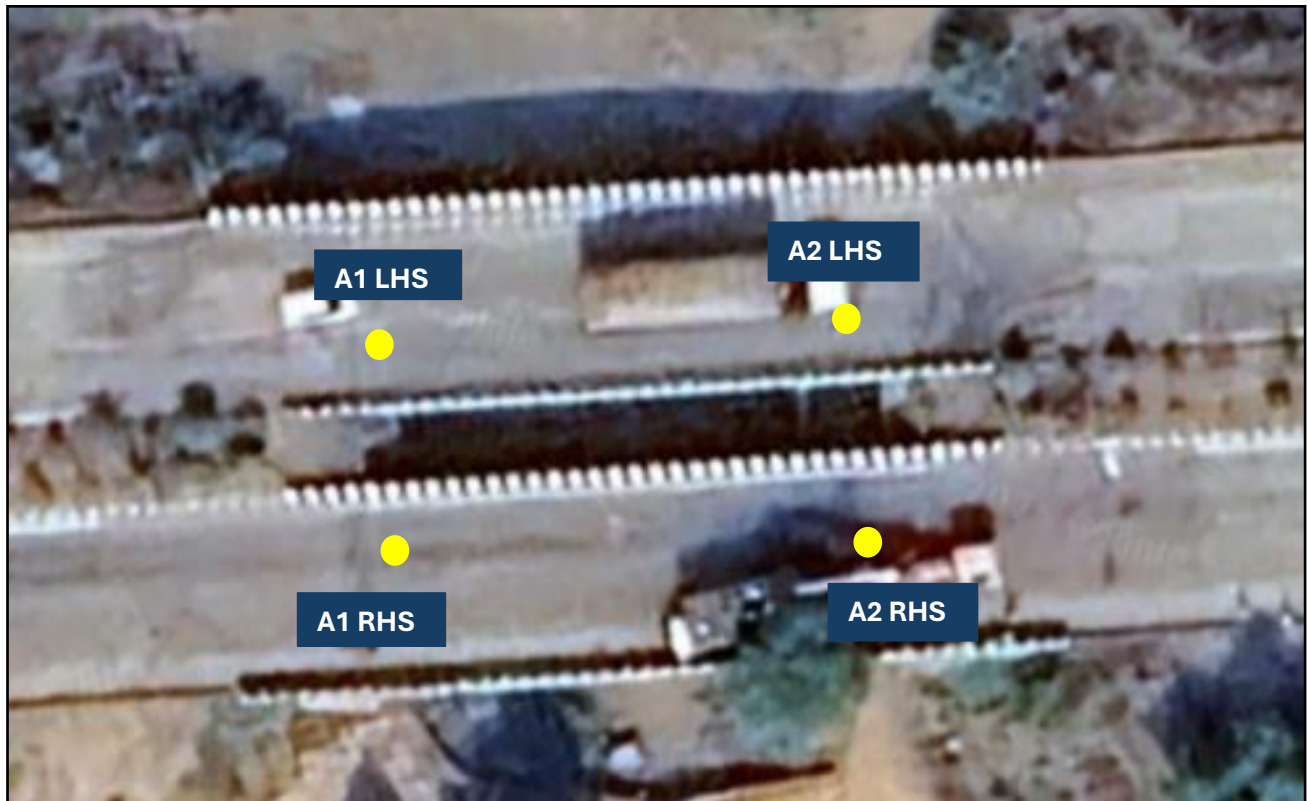


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

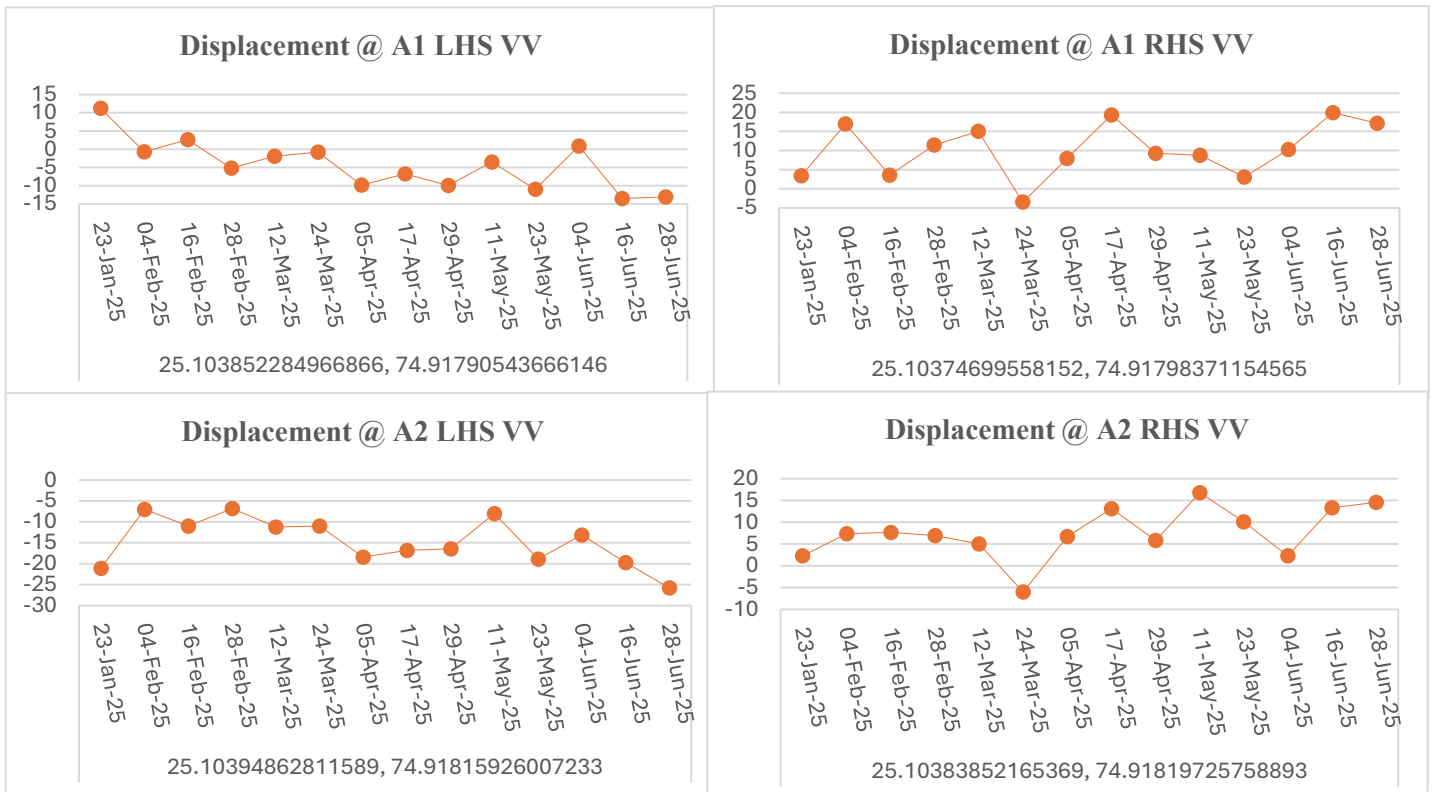
MNB – CH-946+350

Length – 42.89M

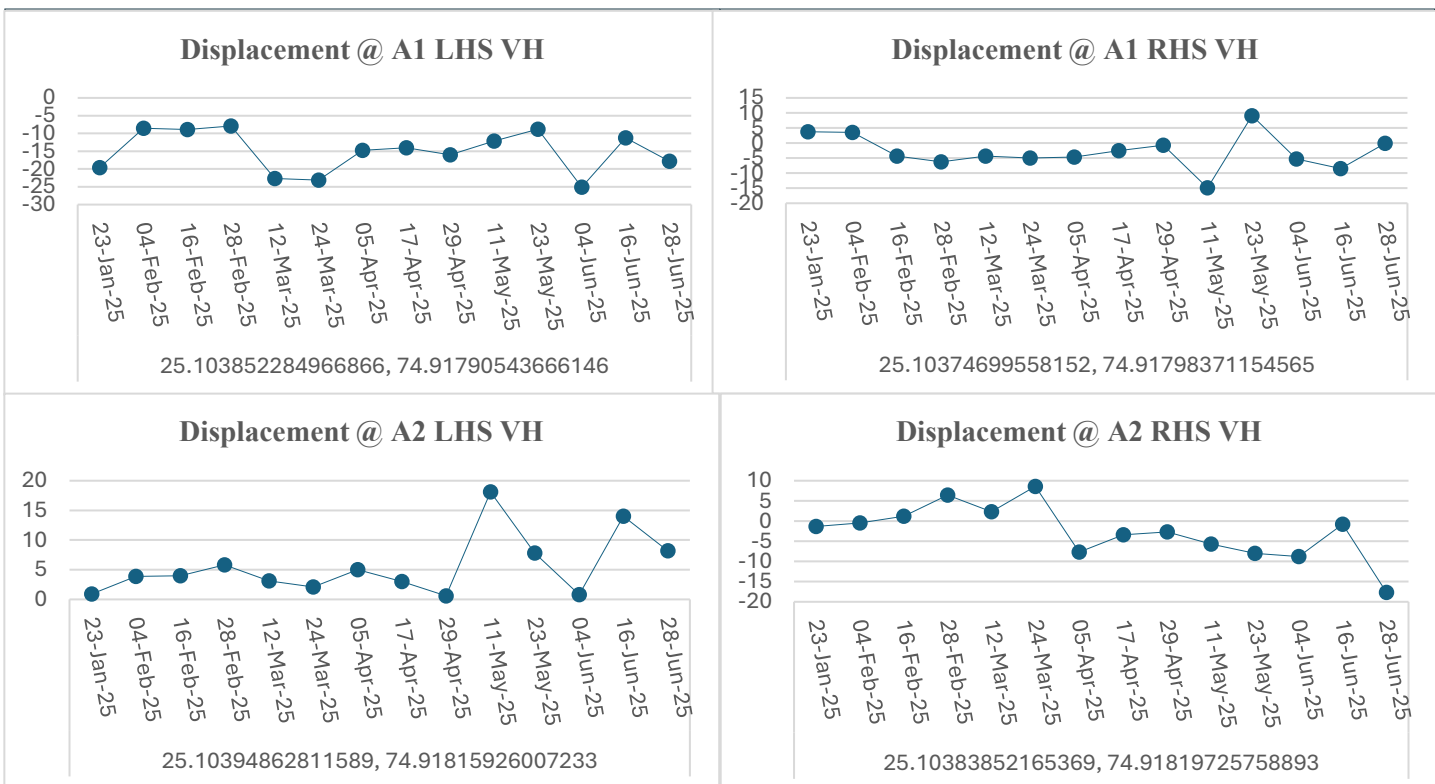


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



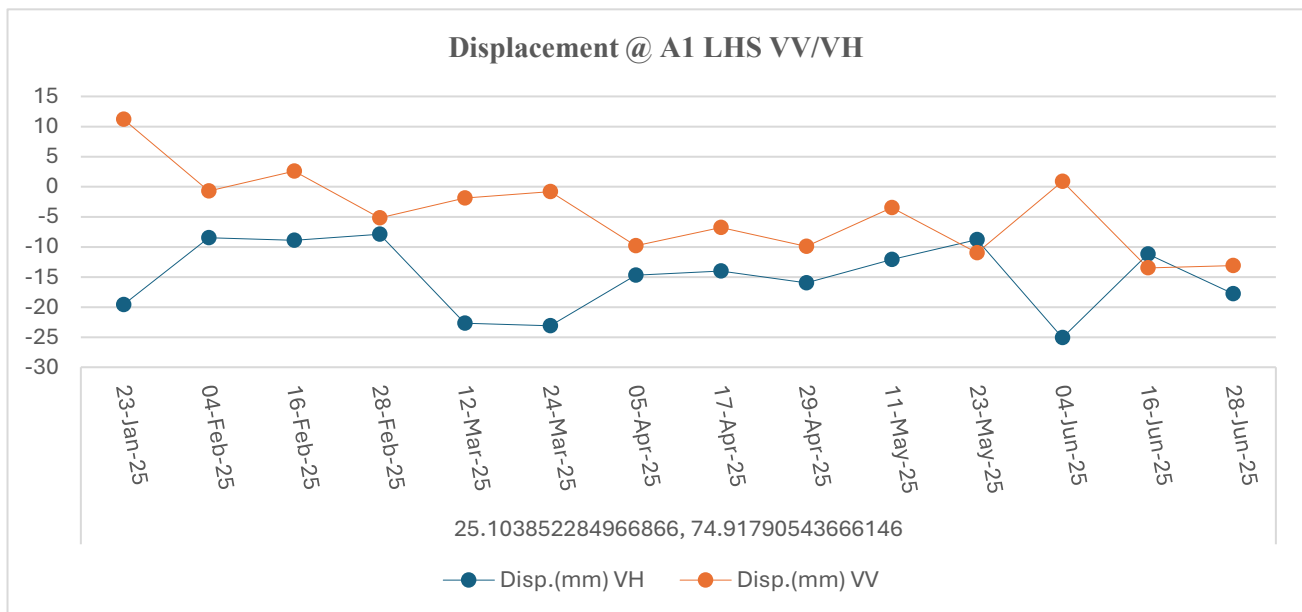
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.103852284966866, 74.91790543666146)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows a generally negative trend throughout the monitoring period, ranging from about -5 mm to -25 mm, indicating consistent downward settlement with periodic fluctuations.
- **Horizontal Displacement (VH)** remains mostly negative as well, fluctuating between -5 mm and -25 mm, suggesting sustained lateral movement toward the south with notable variation across months.

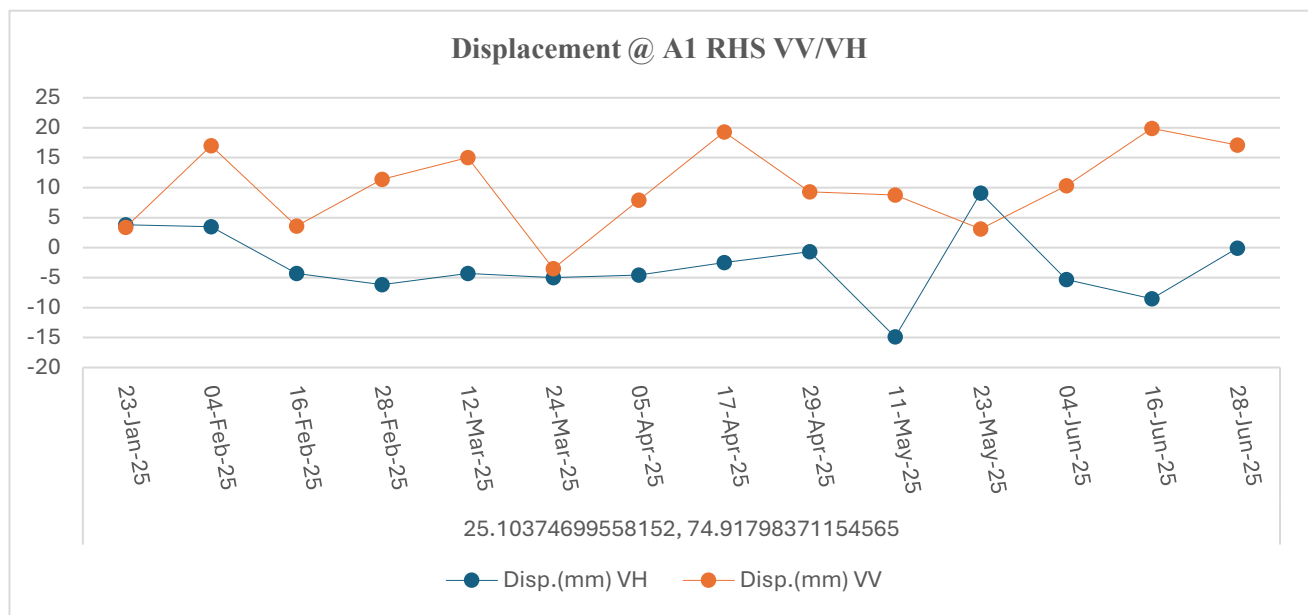
Critical Events & Anomalies:

- **February 2025:** VV and VH both show deeper downward shifts (VV around -10 mm; VH around -15 mm), marking an early period of increased deformation.
- **March 2025:** VH reaches one of its lowest levels (-22 to -25 mm), signalling a significant lateral displacement episode, while VV also dips below -15 mm.
- **June 2025:** Another sharp downward spike occurs in VH (around -25 mm), representing the largest lateral negative displacement in the recorded period, accompanied by VV settling near -10 mm.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.10374699558152, 74.91798371154565)



Key Observations:

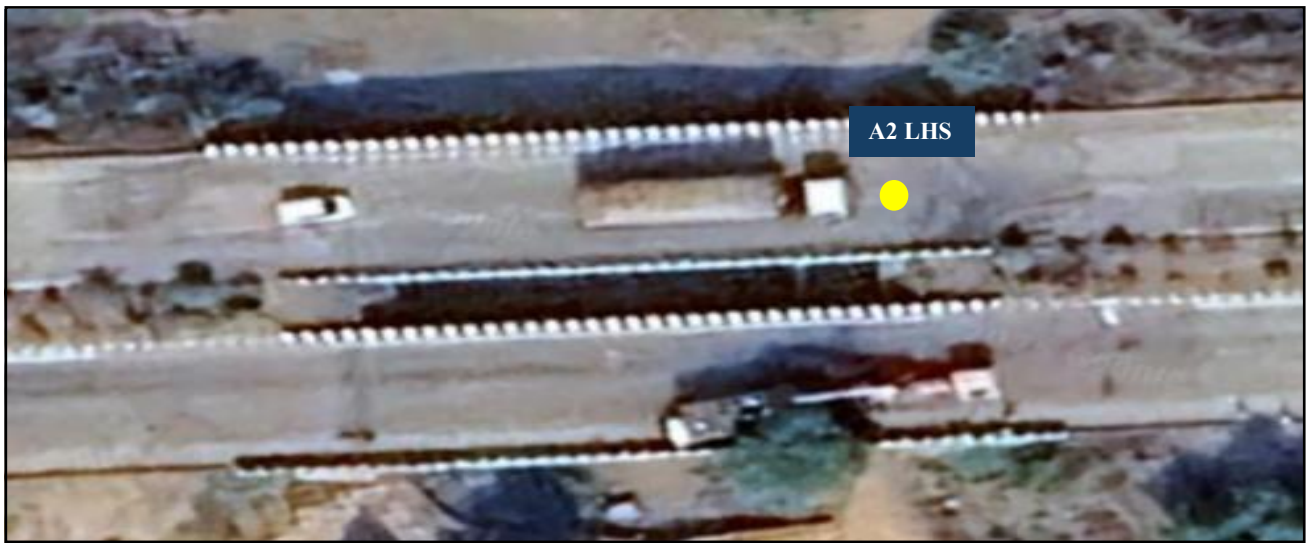
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** fluctuates more widely, ranging from approximately –15 mm to +10 mm, showing alternating cycles of southward and northward lateral movement, with noticeable reversals in direction.
- **Vertical displacement (VV)** shows a predominantly positive trend, generally ranging from +5 mm up to +22 mm, indicating recurring upward vertical movement with several pronounced peaks throughout the monitoring period.

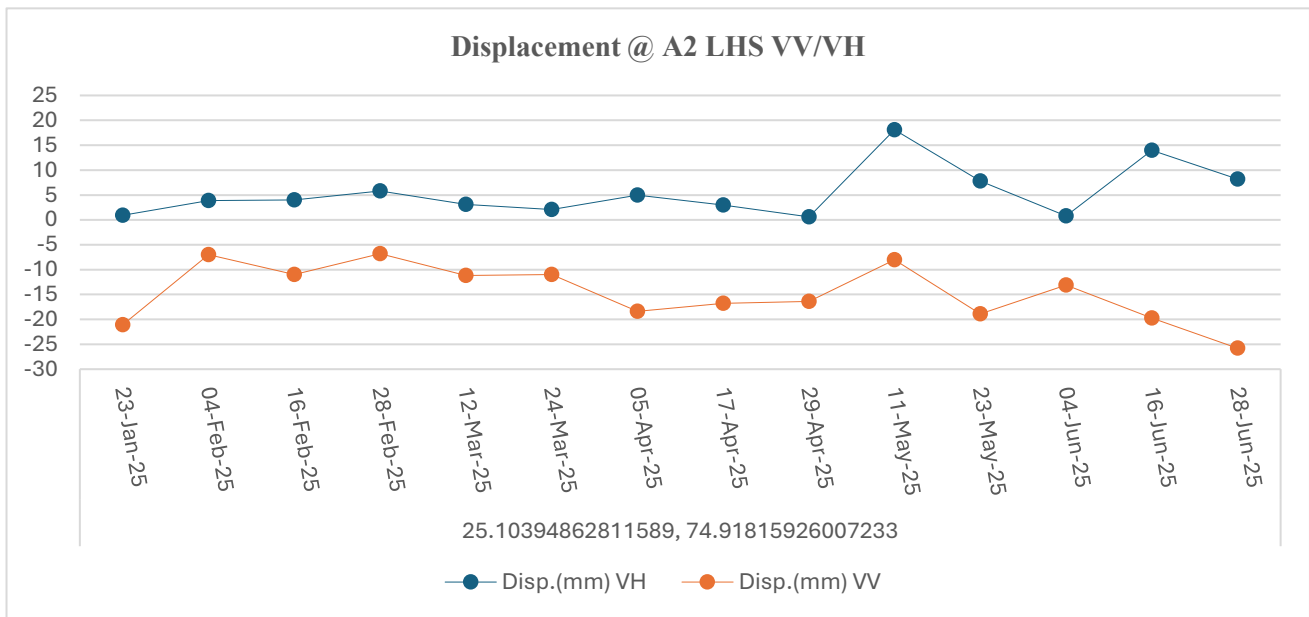
Critical Events & Anomalies:

- **February 2025:** VV spikes to around +18–20 mm, marking the first significant upward movement, while VH trends downward (–5 to –10 mm), indicating opposite vertical–horizontal behaviour at this stage.
- **April–May 2025:** VV reaches another strong peak near +20 mm, while VH hits its lowest point (–15 mm), representing the most pronounced combined uplift and lateral shift in the monitoring period.
- **June 2025:** VV again rises above +18 mm, while VH recovers from earlier lows, returning to moderate positive values (+5 mm), suggesting partial reversal and stabilization in lateral displacement.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.10394862811589, 74.91815926007233)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH trends mostly positive, generally between +2 mm and +18 mm, showing a steady lateral shift over time. A pronounced peak occurs in early May (+18 mm), representing the strongest horizontal movement at this location.
- **Vertical Displacement (VV)** remains consistently negative throughout the monitoring period, ranging from approximately -10 mm to -28 mm. This indicates persistent downward settlement, with fluctuations suggesting cycles of minor recovery followed by renewed downward movement.

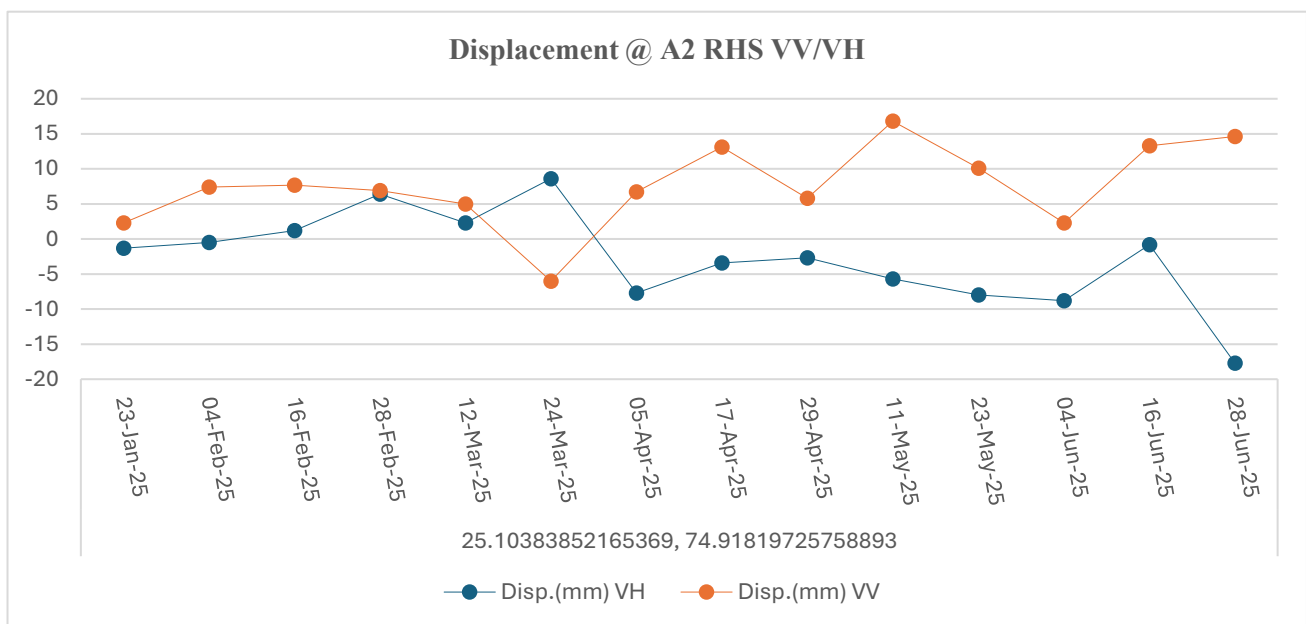
Critical Events & Anomalies:

- **January-February 2025:** VV initially drops sharply to around -20 mm, indicating early settlement acceleration.
- **May 2025:** VH exhibits a significant spike to +18 mm, indicating the highest lateral displacement observed.
- **June 2025:** VV again deepens toward -25 to -28 mm, marking renewed downward settlement.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.10383852165369, 74.91819725758893)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV)** shows a mix of upward and downward movements, generally ranging from about – 10 mm to +18 mm.
- **Horizontal displacement (VH)** displays moderate fluctuations, staying mostly between –5 mm and +10 mm, except for a sharp drop near late June 2025 (–15 mm).

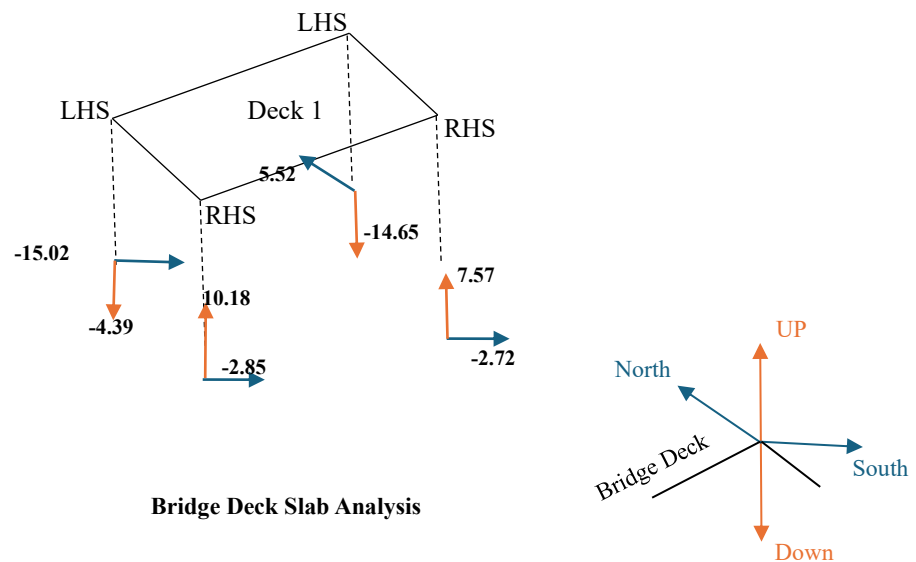
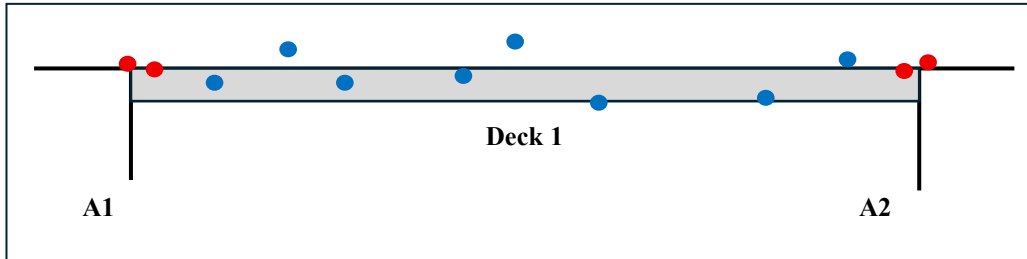
Critical Events & Anomalies:

- **February–March 2025:** VH remains mildly positive (0–5 mm), indicating small but stable northward lateral movement. VV fluctuates between +5 mm and +10 mm, marking a period of mild uplift.
- **May 2025:** VV reaches its peak upward displacement (+18 mm), representing the highest uplift event in the monitoring period.
- **June 2025:** VH experiences a notable downward spike to approximately –15 mm, the largest horizontal deviation on record.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The deck slab shows significant variation in vertical response, with downward movements reaching approximately -15.02 mm to -14.65 mm at multiple RHS and LHS support regions. The combination of high downward displacement on outer edges and localized upward movement near the centre suggests uneven load redistribution along the deck span.

Horizontal Displacement Behaviour:

- Horizontal displacements vary across the deck, with LHS locations showing westward (negative) shifts of about -4.39 mm, while certain RHS zones show eastward shifts up to -2.72 mm. Overall, the pattern reflects non-symmetric horizontal deformation, influenced by both geometry and loading eccentricities.

CONCLUSION:

- The combined vertical (uplift and settlement) and horizontal (lateral shifts in opposing directions) displacement pattern suggests the deck is undergoing torsional distortion rather than uniform bending. The presence of opposite-direction horizontal movements and contrasting vertical responses between LHS and RHS supports confirms deck twisting under load.